Team: **X-Force** | SDC Hackathon 2026 | Domain 2

SURPLUS LINK

B2B Near-Expiry Ingredient Rescue for Malaysian Micro-SMEs

12 RESPONSIBLE CONSUMPTION AND PRODUCTION

13 CLIMATE ACTION

8 DECENT WORK AND ECONOMIC GROWTH

RM1,400 WASTED BETWEEN TWO NEIGHBOURS EVERY WEEK

The Seller's Loss

The Buyer's Overpayment

The Raw Ingredient Waste Problem

RM1,400 WASTED BETWEEN TWO NEIGHBOURS EVERY WEEK

The Seller's Loss

Ahmad bought 20 kg of Arabica beans at RM75/kg, a **RM1,500** order for a corporate event. The event collapsed. He used 8 kg. The remaining 12 kg expires in 17 days. He posted in 3 WhatsApp groups but zero replies. On Friday, 12 kg and ~RM900 went straight into the bin. **0% recovered**.

RM1,400 WASTED BETWEEN TWO NEIGHBOURS EVERY WEEK

The Buyer's Overpayment

1.6 km away, Sarah needs 6 kg of coffee for 50 Tiramisu boxes due Saturday. Her supplier's lead time is 3 days. She drives to Jaya Grocer and pays RM85/kg retail instead of RM75/kg wholesale.
Total: **RM510** instead of **RM450**. That's **~RM60** overpaid plus 3 hours wasted sourcing.

RM1,400 WASTED BETWEEN TWO NEIGHBOURS EVERY WEEK

The Raw Ingredient Waste Problem

That's RM900 destroyed + RM510 overpaid = **RM1410** lost between two neighbours in one week.

Malaysian F&B businesses waste raw ingredients through spoilage and overstocking. A study of 15 restaurants recorded **2,186 kg of food waste** in one week, with **51.37%** from preparation losses, **including spoiled raw materials**.

(Source: Environmental Monitoring and Assessment, 2023)

WHO WE SERVE

Sellers

Micro-café, bakery, or F&B owner with surplus bulk ingredients approaching Best Before date. Typically 1–5 staff, revenue RM15,000–30,000/month.

Buyers

Nearby micro-F&B business needing the same ingredient for immediate production (within 24–72 hours).

WHO WE SERVE

Why Them:

They currently use WhatsApp groups and Facebook to offload surplus unstructured, unsearchable, no trust, no pricing logic.

Not Our Users:

Large restaurant chains (they have procurement teams).
Consumers - usually have buy small amounts of ingredients.

THE SOLUTION

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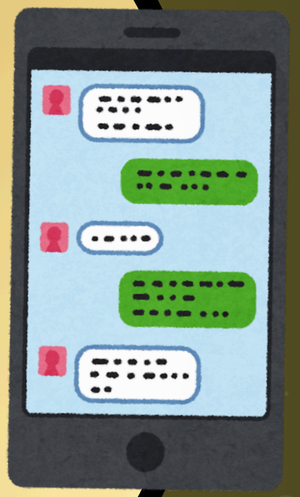
AI-Assisted Listing • Smart Matching • Human-Controlled Deals

THE CORE JOURNEY



TRIGGER

Ahmad doesn't fill out a form. He just types what he'd normally say:
"**12kg arabica unopened exp nov 12 paid rm75**"

AI extracts: product name, quantity, unit, original cost, expiry date, condition.

Hard rule, not AI judgment: **no expiry date/photo proof = listing cannot publish.**

Ahmad reviews the AI-drafted listing, edits if needed, sets asking price.

THE CORE JOURNEY

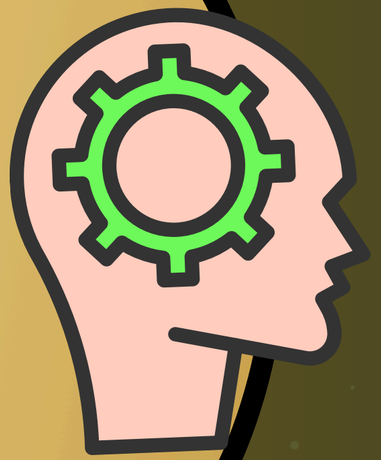
CONNECTED DATA (AI MATCHING)

AI **scans nearby buyer profiles** for who actually needs this ingredient based on their menu items and urgent needs, not just location.

Sarah sees it under "Recommended for You," not buried in a random Facebook feed

Listing shows: **product, price, expiry, distance, seller rating** everything Sarah needs to decide in 5 seconds

THE CORE JOURNEY



DECISION LOGIC (CHAT & BARGAIN)

Sarah messages Ahmad directly and **asks questions, negotiates price, confirms condition.**

AI does **NOT negotiate**. It can **only summarize**: "You both agreed on 6kg at RM45/kg.".

Keeps the **human trust** and back-and-forth real business relationships are built on.

THE CORE JOURNEY

HUMAN APPROVAL (DEAL CARD)

Either party clicks [Generate Deal Card] and locks in quantity, price, delivery method, handover time/location

Both Ahmad and Sarah must confirm the deal before anything is final.

Once confirmed: stock is **reserved automatically**, preventing Ahmad from double-selling the same 12kg

THE CORE JOURNEY

VISIBLE OUTCOME (HANDOVER & FEEDBACK)

Sarah collects the beans, clicks [Confirm Goods Received]

Two things happen immediately:

- Private feedback to AI (was the match accurate? was it as described?) — improves future matching, never shown publicly
- Public rating (1–10, Shopee-style) + review tags — builds Ahmad's trust score for future buyers

TRUST, SAFETY & CONTROL

No expiry proof = no listing — enforced before publishing, not a courtesy.

AI never penalizes. If a buyer reports an issue, AI compares evidence side-by-side, but a human moderator makes the final call to warn, suspend, or ban.

Every listing, chat, deal, and dispute is logged in a full audit trail, nothing happens invisibly.

WHERE AI FITS

AI is involved in exactly four places — no more, no less:

AI Function	What It Actually Does
Listing Assistant	Converts messy text/photos into a structured listing.
Price Advisor	Suggests a fair discount based on days-to-expiry and category.
Smart Matching	Finds the nearby buyer most likely to need this item.
Chat Summarizer	Extracts agreed terms from a conversation and it never negotiates

WHERE HUMANS & RULES STAY IN CONTROL

Everything that actually decides something stays outside AI's hands:

AI Function	What It Actually Does
Expiry Gate	Hard block — no proof, no listing, no exceptions.
Deal Card Lock	Requires explicit confirmation from both people.
Stock Reservation	Simple system logic — prevents double-selling.
Dispute Resolution	Human moderator makes the final penalty call.
Payment & Pickup Record	Logged, not processed — users arrange it themselves.

BUSINESS IMPACT (EXPANDED)

The Ahmad & Sarah Deal, Solved:

Before SurplusLink:

Ahmad loses RM900, Sarah overpays RM60 + 3 hours,
RM1,410 wasted between two neighbours in one week.

After SurplusLink:

Ahmad sells 6kg at RM45/kg (~RM270), Sarah pays RM270 instead of RM510 retail — saves **RM240** + the **3 Hours**.

Combined value recovered in this single transaction: **~RM510**

BUSINESS IMPACT (EXPANDED)

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RESPONSIBLE CONSUMPTION AND PRODUCTION

Target 12.3: halve food waste. SurplusLink directly prevents edible surplus from being discarded.

DECENT WORK AND ECONOMIC GROWTH

Target 8.3: supports micro-enterprise viability. Protects thin SME margins by recovering lost inventory cost.

CLIMATE ACTION

food waste in landfills produces methane; ~2kg CO_2e avoided per 1kg of food saved (FAO estimate).

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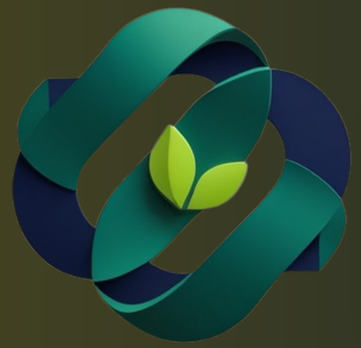

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Thank you